22-8-24 EX:7 SLIDING WINDOW PROTOCOL

AIM:

To write a pgm to implement flow control at data link layer using sliding window protocol simulate flow of frames from one node to another.

Sender code:

```
import time
import os
print("="*55)
print("SLIDING WINDOW PROTOCOL - SENDER")
window_size = int(input("Enter window size:"))
if window_size <=0:
    print("Invalid window size:")
    exit()
message = input("Enter text message")
if not message:
    print("message cannot be empty!")
    exit()
frames = []
for i, character in enumerate(message):
    frames.append({"frame no":i, "data":
    character})
print("\n Frames created:")
for frame in frames:
    print("Frame NO: {frame["frame_no"]}
DATA: {frame['data']}")
total_frame = len(frames)
base=0
```

Sender code:

Frame NO: 0 DATA: H

Frame NO: 1 DATA: E

Frame NO: 2 DATA: L

Waiting for acknowledgement

Receiver response: ACK

ACK is as expected

Sliding window moves forward

Sender - Buffer:

Frame NO: 3 DATA: L

Frame NO: 4 DATA: O

Waiting for acknowledgement

Receiver response: ACK5

ACK is as expected

Sliding window moves forward.

original message: HELLO

Receiver code:

```
import os
SENDER_FILE = "sender_buffer.txt"
RECIEVER_FILE = "receiver-buffer.txt"
expected_frame=0
while True:
    if not os.path.exists(SENDER_FILE):
        print("sender_buffer not found")
        break
    with open(SENDER_FILE,"r") as file:
        lines = file.readlines()
    if len(lines)==0:
        print("NO frames available")
```

```
break
print("\n Frames received")
error = False
error_frame = None
try:
    ack_number = int(charto[i])
except ValueError:
    print("Invalid ACK/NACK number!")
    continue
print("\n receiver response : {ack_type}{ack_number}")
if ack_type == "ACK":
    if ack_number == end:
        print("ACK is as expected")
        print("sliding : window moves forward")
        base = ack_number
    elif base < ack_number <= end:
        print("cumulative Ack received")
        print("Moving window to from {ack_number}")
        base = ack_number
else:
    print("unexpected ACK")
    print("resending current window")
elif ack_type == "NACK":
        print("NACK received from frame:
        {ack_number}")
    print("retransmitting frames from that frame")
```

```
if (base <= ack_number < end):
    base = ack_number
else:
    print("Invalid NACK number")
    print("Resending current window")
else:
    print("Invalid response received")
    time.sleep(1)
print("All frames transmitted successfully!")
print("original message: {message}")
```

O/P:

```
SLIDING WINDOW PROTOCOL SENDER
Enter window size: 3
Enter test message: HELLO
```

RESULT:

Thus the sliding window protocol was implemented successfully using python & the flow of frames the sender and receiver was simulated using ACK/NACK.